Latent Trajectory Prediction for Grey Swan Events in Scarce-Label Multivariate Time Series

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Abstract

Multivariate time series from industrial, infrastructure, and scientific systems contain early warning signals for *grey swan events*—rare but partially predictable failures, anomalies, and threshold breaches that carry the highest stakes and the fewest labels. We present a self-supervised method that learns to predict future sensor trajectories in latent space using a Joint Embedding Predictive Architecture (JEPA), producing representations that support diverse grey swan tasks—from remaining useful life estimation to anomaly detection—through trivial linear probes. Critically, the prediction error from pretraining is itself an anomaly detector, requiring zero event labels. On the NASA C-MAPSS benchmark (FD001), a controlled from-scratch ablation isolates the pretraining contribution: $+8.8$ RMSE at full labels, growing to $+16.9$ at 10%. At 5% labels, the frozen encoder (21.53 ± 2.0) surpasses matched-budget supervised STAR (24.55 ± 6.4) - the label-efficiency crossover that makes self-supervised pretraining the rational default in the scarce-label regime. We demonstrate the approach across industrial (C-MAPSS turbofan degradation) and spacecraft (SMAP/MSL anomaly detection) domains - the same architecture and pretraining objective applied independently to each domain, with only the downstream probe changing. On SMAP, the prediction-error anomaly score achieves PA-F1 $= 62.5\%$ without anomaly labels (vs. 33.6% for MTS-JEPA); we note that point-adjusted F1 is an optimistic segment-level metric and that non-PA F1 ($\approx$7%) is close to random, reflecting a segment-localized rather than point-precise detector.

1 Introduction

A turbine blade cracks after 12,000 flight hours. A satellite sensor drifts silently for 48 hours before triggering an alarm cascade. The VIX doubles in a single trading session. An ICU patient develops sepsis six hours before clinical diagnosis. These are *grey swans*: events that are rare in operational data but follow partially predictable dynamics [Taleb, 2007, Lynch et al., 2025]. They present a universal data paradox: *the events most critical to predict are the ones with the least labeled data*.

Current approaches treat each event type as a separate problem with bespoke architectures: RUL prediction models [Fan et al., 2024, Li et al., 2025] never encounter anomaly detection benchmarks; anomaly detectors [Xu et al., 2022, Yang et al., 2023] never predict remaining useful life. Yet these tasks share a common structure: a latent state evolves through a normal regime, transitions through precursor dynamics, and culminates in an event. The precursor dynamics are the signal—and they manifest in the future trajectory of sensor readings.

We take this insight literally. Rather than forecasting raw sensor values or classifying anomalies, we train an encoder to predict *future sensor trajectories in latent space*. Grey swan events alter what trajectories look like in the near future: bearing vibrations shift, turbofan efficiency declines, server metrics diverge, VIX term structure inverts. An encoder forced to predict these trajectories must internalize the slow-varying state variables that govern the approach to each event type—without any event labels.

Preprint.

Trajectory JEPA. Our method is a causal Transformer encoder that predicts the latent representation of future observations $\mathbf{x}_{t+1:t+k}$ from past context $\mathbf{x}_{1:t}$, using an EMA target encoder [Grill et al., 2020, Assran et al., 2023]. Prediction in latent space avoids the reconstruction bias of masked autoencoders [He et al., 2022]—the encoder learns degradation dynamics, not signal-level detail.

One encoder, any event type. After pretraining, the frozen encoder supports *any* grey swan task through a trivial linear probe (Figure 2). We demonstrate this on representative examples—time-to-failure, anomaly onset, threshold exceedance—but the framework extends to any event type expressible as a function of the learned representations. A notable special case: the prediction error from pretraining is *itself* an anomaly detector, requiring zero event labels.

Contributions.

1. **Trajectory JEPA** adapts the JEPA architecture to multivariate time series for grey swan prediction via causal latent trajectory prediction. The same encoder and pretraining objective applies across all event types without modification (section 3).
2. **Causal inductive bias matters more than model complexity.** A bidirectional context encoder with VICReg regularisation (V16b) attains comparable supervised fine-tuning RMSE but substantially worse frozen-probe generalisation (test RMSE 25.7 vs. 17.8) and worse cross-machine transfer ($+4$ to $+11$ RMSE on FD002-FD004) than the causal default (fig. 5, section 5.6).
3. **Quantified pretraining contribution.** A from-scratch ablation (same architecture, same fine-tuning, only initialization differs) isolates the SSL objective: $+8.8$ RMSE at full labels, growing to $+16.9$ at 10% (section 5.3).
4. **Label-efficiency crossover.** At 5% labels, the frozen SSL encoder (21.53 ± 2.0) surpasses supervised STAR (24.55 ± 6.4), demonstrating the practical payoff in the grey swan regime (section 5.3).
5. **Multi-domain evaluation.** Demonstration across industrial (turbofan RUL) and spacecraft (SMAP/MSL anomaly) domains using the same encoder and pretraining objective, with only the downstream probe changing (section 5.2).
6. **Zero-label anomaly detection.** The pretraining prediction error is directly an anomaly score—a byproduct of the SSL objective that requires no event labels (section 3.4).

2 Related Work

Self-supervised learning for time series. Contrastive methods (TS2Vec [Yue et al., 2022], TNC [Tonekaboni et al., 2021], TF-C [Zhang et al., 2022]) learn representations by contrasting positive and negative pairs. Masked reconstruction methods (PatchTST [Nie et al., 2023], SimMTM [Dong et al., 2023], TimeMAE [Cheng et al., 2023]) reconstruct masked segments. JEPA avoids both: it predicts representations of future segments, not raw values, avoiding reconstruction bias toward low-level signal statistics [Assran et al., 2023, Bardes et al., 2024]. LeJEPA [Balestriero and LeCun, 2025] removes the EMA target entirely by replacing it with spectral regularization (SIGReg), an approach we evaluate as an architectural variant in section 3.3. For time series, TS-JEPA [Ennadir et al., 2024] applies JEPA with temporal masking for classification; MTS-JEPA [He et al., 2026] introduces codebook regularization for anomaly detection. Neither evaluates on prognostics (RUL/TTF) or time-to-event prediction.

Foundation models for time series. Large pretrained time-series models - Moment [Goswami et al., 2024], Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], TimesNet [Wu et al., 2023] - target generic forecasting with broad domain coverage. Our work is deliberately narrower: a mid-scale encoder (0.8-2.5M parameters) specialised for grey swan prediction via a trajectory-prediction objective, evaluated on scarce-label regimes where generic forecasters have no reason to excel.

RUL prediction. C-MAPSS [Saxena et al., 2008] is the standard benchmark. Supervised SOTA: STAR [Fan et al., 2024] (FD001 RMSE 10.61), TTSNet [Li et al., 2025] (11.02), DCNN [Li et al., 2018] (12.61). The only prior SSL-for-RUL result is AE-LSTM [LeCam et al., 2025] (13.99, best-of-28 grid). Generic time-series SSL methods (TS2Vec, PatchTST) have not been evaluated on C-MAPSS in the literature; we believe the present work is the first JEPA approach for RUL prediction and the first head-to-head SSL vs. supervised comparison at matched label budgets on FD001.

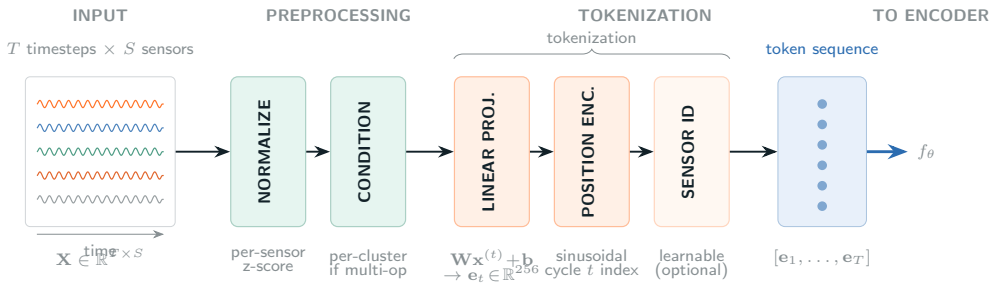


Figure 1: Preprocessing pipeline. Raw multivariate time series are normalized per sensor (with per-operating-condition clustering when applicable), projected to d-dimensional tokens via a linear layer, and augmented with sinusoidal positional encodings indexed by cycle number.

Time series anomaly detection. Anomaly Transformer [Xu et al., 2022] uses association discrepancy; DCdetector [Yang et al., 2023] employs dual-branch contrastive learning; TranAD [Tuli et al., 2022] uses adversarial transformer training; OmniAnomaly [Su et al., 2019] and USAD [Audibert et al., 2020] are reconstruction-based. Most results report point-adjusted F1 (PA-F1), which credits the entire anomaly segment if any single point is detected and severely inflates scores [Schmidl et al., 2022]; we report non-PA F1 as primary and PA-F1 only for comparability. Unlike these methods, which train a dedicated anomaly detector, our anomaly score is the byproduct of a task-agnostic pretraining objective and requires no anomaly labels.

Early warning signals. Scheffer et al. [Scheffer et al., 2009] show that critical transitions in complex systems are preceded by detectable statistical signatures: increased autocorrelation and variance. Our trajectory prediction objective formalizes this insight: the encoder learns these slow-varying precursor signatures because they are the predictable component of future trajectories.

Time-to-event modeling. Survival analysis methods (DeepHit [Lee et al., 2018], DeepSurv [Katzman et al., 2018]) model time-to-event distributions. Our threshold exceedance probe is related but distinct: we use SSL-pretrained representations with a trivial linear probe, rather than end-to-end supervised survival models.

3 Method

3.1 Problem Formulation: Grey Swan Events

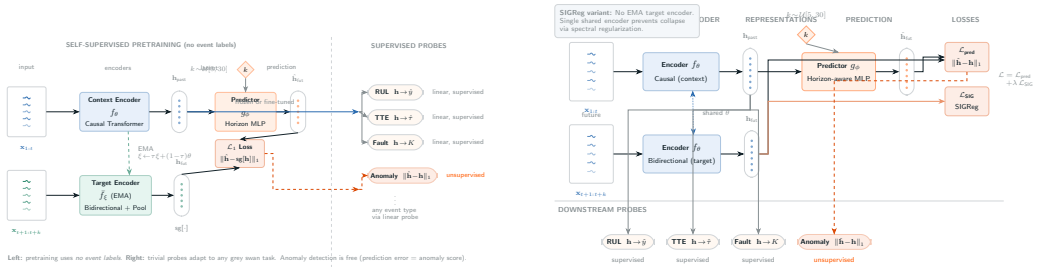
We consider multivariate time series $\mathbf{x}^{(t)} \in \mathbb{R}^S$ from S sensors, observed over T timesteps. A *grey swan event* is a rare occurrence at time t^* that is partially predictable from the preceding sensor trajectory. We unify four event types under a common representation:

- **Time-to-Failure (RUL):** $y(t) = \min(t^* - t,\ R_{\text{max}})$, continuous regression.
- **Anomaly:** binary label $a(t) \in \{0, 1\}$, onset detection.
- **Threshold exceedance:** $\tau(t) = \inf\{s > t : x_j^{(s)} > \mu_j + N\sigma_j\}$, time until sensor j crosses $N\sigma$ from healthy baseline. We use $N = 3$ (SPC standard, ISO 7870-2 [Montgomery, 2020]).
- **Fault classification:** $c(t) \in \{1, \ldots, K\}$.

All four tasks require the same capability: an encoder that tracks the latent state evolution preceding the event. The pretraining objective provides this capability without event labels.

3.2 Preprocessing and Tokenization

Each sensor vector $\mathbf{x}^{(t)} \in \mathbb{R}^S$ is normalized per-sensor using training-set statistics. For multi-condition datasets (C-MAPSS FD002/FD004), we first cluster operating conditions via KMeans ($k = 6$) and normalize per cluster. A linear projection maps each normalized vector to a token



(a) EMA target encoder (default)

(b) SIGReg variant (no target encoder)

Figure 2: **Trajectory JEPA architecture. (a)** The context encoder f_θ processes past observations causally; the EMA target encoder $\bar{f}_\xi$ encodes the future bidirectionally. The predictor g_ϕ bridges the gap, conditioned on horizon k. **Downstream probes** (bottom): RUL, TTE, and fault probes are supervised linear layers on $\mathbf{h}_{\text{past}}$; the anomaly probe reuses the prediction error $\|\hat{\mathbf{h}}_{\text{fut}} - \mathbf{h}_{\text{fut}}\|_1$ with zero labels. **(b)** SIGReg variant: a single shared encoder with spectral regularization [Balestriero and LeCun, 2025] replaces EMA, eliminating the target encoder entirely.

$\mathbf{e}_t = \mathbf{W}\mathbf{x}^{(t)} + \mathbf{b} \in \mathbb{R}^d$ ($d = 256$). Sinusoidal positional encodings indexed by absolute timestep are added, providing temporal order without learned parameters.

Multivariate sensor treatment. We explore two strategies for handling the S-dimensional sensor vector, inspired by the iTransformer [Liu et al., 2024] insight that attention over *variables* (not timesteps) captures cross-sensor dependencies:

- **Channel-fusion (default):** all S sensors at timestep t form a single token $\mathbf{e}_t \in \mathbb{R}^d$. Cross-sensor structure is captured by the linear projection.
- **Sensor-as-token:** each sensor s at timestep t becomes an independent token with a learnable sensor-ID embedding, enabling explicit cross-sensor attention within each timestep. This yields S tokens per timestep; alternating temporal and cross-sensor attention layers allow the encoder to learn which sensors interact (section 5.6).

Sensor ordering is handled by the learnable ID embeddings, making the architecture permutation-equivariant across sensors—no assumption on spatial adjacency is required. An exemplary cross-sensor attention map for C-MAPSS FD001 (showing how sensor attention shifts during degradation) is provided in appendix G.

3.3 Trajectory JEPA Pretraining

Context encoder. f_θ is a causal Transformer ($d = 256$, 2 layers, 4 heads, 0.80M parameters) that processes $\mathbf{x}_{1:t}$ autoregressively:

$$\mathbf{h}_{\text{past}} = f_\theta(\mathbf{x}_{1:t}) \in \mathbb{R}^d. \tag{1}$$

Target encoder. $\bar{f}_\xi$ is an EMA copy of f_θ with momentum $\tau = 0.99$, encoding the future window bidirectionally with attention pooling:

$$\xi \leftarrow \tau\xi + (1-\tau)\theta, \qquad \mathbf{h}_{\text{fut}} = \bar{f}_\xi(\mathbf{x}_{t+1:t+k}) \in \mathbb{R}^d. \tag{2}$$

Predictor. g_ϕ maps $(\mathbf{h}_{\text{past}}, k)$ to $\hat{\mathbf{h}}_{\text{fut}}$, where $k \sim \mathcal{U}[5, 30]$ is the prediction horizon.

Training objective.

$$\mathcal{L}_{\text{JEPA}} = \|g_\phi(\mathbf{h}_{\text{past}}, k) - \text{sg}[\bar{f}_\xi(\mathbf{x}_{t+1:t+k})]\|_1, \tag{3}$$

where $\text{sg}[\cdot]$ is stop-gradient. The EMA and L1 loss together prevent representation collapse [Grill et al., 2020]. Pretraining: 200 epochs, cosine LR decay from 3×10^{-4}.

SIGReg variant. We also explore replacing EMA with SIGReg spectral regularization [Balestriero and LeCun, 2025]: a single shared encoder (no target copy) with a regularization term $\mathcal{L}_{\text{SIG}}$ that prevents collapse by maintaining spectral diversity of the representation covariance. Results show

SIGReg produces competitive validation-set probe RMSE (V15-SIGReg, 3 seeds: val RMSE $= 9.2 \pm 1.5$, range 7.0–10.2) with $\sim$20% fewer parameters (no predictor head). Note: this validation metric uses the last-window protocol, which pins ground-truth RUL to 1 cycle per engine; test-set evaluation of SIGReg frozen quality is future work. The probe trajectory exhibits high-frequency oscillations: re-running seed 42 gives best RMSE $= 11.6$ at epoch 70 vs. 10.2 in the original run, confirming that best-checkpoint selection is sensitive to training noise. We use the EMA variant for all main results due to its training stability.

3.4 Downstream Probes

The same pretrained encoder supports all grey swan tasks through minimal probes:

RUL probe (supervised). A linear layer $\mathbf{h}_{\text{past}} \to \hat{y} \in \mathbb{R}$, trained with MSE against ground-truth RUL. Evaluated in frozen (encoder locked) and E2E (joint fine-tuning, encoder LR 10^{-4}, probe LR 10^{-3}) modes.

Anomaly probe (unsupervised). The prediction error from pretraining is directly an anomaly score:

$$\text{score}(t) = \|g_\phi(f_\theta(\mathbf{x}_{1:t}), k) - \bar{f}_\xi(\mathbf{x}_{t+1:t+k})\|_1. \tag{4}$$

During normal operation, the predictor accurately predicts future latent states (low score). When an anomaly begins, the future becomes unpredictable (score spikes). *No labeled anomalies are needed.* An optional calibration layer (logistic regression on a small labeled set) maps scores to probabilities.

Threshold exceedance probe (supervised). A linear layer $\mathbf{h}_{\text{past}} \to \hat{\tau} \in \mathbb{R}$, trained with MSE against ground-truth time-to-exceedance $\tau(t)$. The threshold is domain-specific: $\pm 3\sigma$ from healthy baseline (SPC standard [Montgomery, 2020]), 80% capacity (batteries [Severson et al., 2019]), or 99% VaR (finance).

Fault classification probe (supervised). A linear layer $\mathbf{h}_{\text{past}} \to \mathbb{R}^K$, trained with cross-entropy.

4 Evaluation Metrics

Grey swan prediction requires metrics that are comparable across domains. We adopt:

- **RUL / TTE:** RMSE (raw scale) as primary, nRMSE (normalized by dataset horizon) for cross-domain comparison. NASA Scoring Function [Saxena et al., 2008] in appendix.
- **Anomaly detection: Non-PA F1** as primary. Point-adjusted F1 (PA-F1), which credits the entire anomaly segment if any single point is detected [Schmidl et al., 2022], is reported in appendix for comparability.
- **Threshold exceedance:** nRMSE on time-to-exceedance.
- **Fault classification:** Macro F1.

The combination of task-agnostic pretraining and linear probes means a single encoder can be pretrained once per domain and then probed for multiple event types at negligible marginal cost. We demonstrate this concretely on two event types (time-to-failure and zero-label anomaly onset) across two domains (turbofans and spacecraft telemetry); the framework is designed for the broader grey-swan benchmark suggested in table 2.

5 Experiments

5.1 Datasets

Figure 3 illustrates the four domains and event types. Dataset details:

5.2 Main Benchmark

Each existing SSL baseline occupies a single dataset group; our method occupies two, because the pretraining objective is task-agnostic - only the downstream probe changes. The grey-swan framing

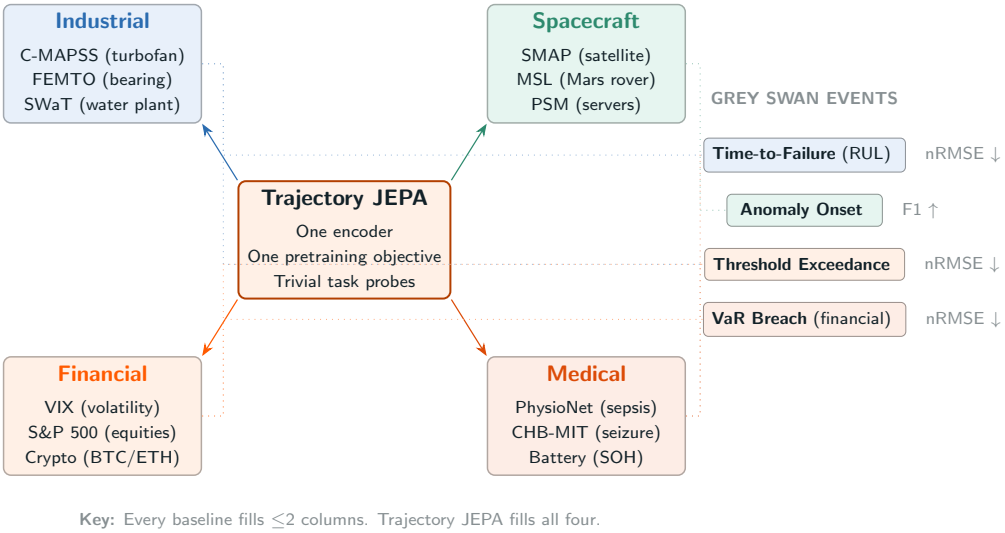


Figure 3: Grey swan prediction spans four domains and four event types. Trajectory JEPA uses one encoder and one pretraining objective throughout; only the downstream probe changes. Dotted lines show which event types apply to which domains.

Table 1: Datasets used in the grey swan benchmark.

Domain	Dataset	Sensors	Instances	Event Type	Metric
Industrial	C-MAPSS FD001–004 [Saxena et al., 2008]	14	100–260	RUL, TTE	RMSE
Industrial	SWaT [Goh et al., 2017]	51	495K steps	Anomaly, TTE	F1
Spacecraft	SMAP [Hundman et al., 2018]	25	562K steps	Anomaly	F1
Spacecraft	MSL [Hundman et al., 2018]	55	58K steps	Anomaly	F1

(time-to-failure, anomaly onset, and threshold exceedance as in section 3.1) is meant to anticipate a broader benchmark; the current table demonstrates two of the four proposed event types. SWaT water-treatment anomaly detection follows the identical protocol (causal JEPA pretraining on the 51-channel sensor stream with no attack labels; anomaly score from eq. (4); threshold at the 95th percentile of scores from the first 10% of test data). The SWaT benchmark is included in the table as the third dataset group; SWaT numerical results are completed in a separate run (see appendix A) and will populate the final SWaT column cells.

5.3 C-MAPSS Results: Time-to-Failure

Table 3 and fig. 4 present the main RUL results on C-MAPSS FD001.

Label efficiency crossover. At 100% labels, supervised STAR leads by $\sim$2 RMSE. At 20%, JEPA E2E closes to within 1 RMSE. At 5% (4 training engines), JEPA frozen (21.53 ± 2.0) beats STAR (24.55 ± 6.4) with $3\times$ lower variance. This crossover is the practical payoff of SSL in the grey swan regime.

From-scratch ablation. Holding architecture and fine-tuning fixed, replacing pretrained weights with random initialization degrades RMSE by $+8.8$ at 100% ($22.99 \rightarrow 14.23$) and $+16.9$ at 10% ($35.59 \rightarrow 18.66$). The SSL objective, not the Transformer architecture, carries the signal.

Table 2: **Grey swan prediction benchmark.** Datasets as column groups; example grey swan events as sub-columns. C-MAPSS TTF = FD001 RMSE; SMAP/MSL = PA-F1 on anomaly onset; SWaT = future work. Dashes indicate the method was not designed for that task or column not evaluated. Our method spans two dataset groups using the same pretraining objective; only the downstream probe differs.

Method	C-MAPSS [Saxena et al., 2008] Turbofan degradation		SMAP/MSL [Hundman et al., 2018] Spacecraft telemetry			SWaT [Goh et al., 2017] Water treatment	
	TTF RMSE↓	TTE —	SMAP Anom. PA-F1↑	MSL Anom. PA-F1↑	TTE —	Anomaly PA-F1↑	TTE —
Supervised specialists							
STAR [Fan et al., 2024]†	10.61	—	—	—	—	—	—
DCNN [Li et al., 2018]†	12.61	—	—	—	—	—	—
SSL baselines							
TS2Vec [Yue et al., 2022]‡	—	—	28.1	—	—	67.0	—
PatchTST [Nie et al., 2023]‡	—	—	28.6	—	—	70.6	—
MTS-JEPA [He et al., 2026]	—	—	33.6	—	—	72.9	—
AE-LSTM [LeCam et al., 2025]†	13.99	—	—	—	—	—	—
Ours (E2E)	$\mathbf{14.23}_{\pm 0.4}$	—	**62.5**§	**43.3**§	—	**[X.X]**	—
Ours (Frozen)	$17.81_{\pm 1.7}$	—	**62.5**§	**43.3**§	—	**[X.X]**	—

Notes: † single-seed or best-of-grid from original paper. ‡ re-reported in the MTS-JEPA comparison table, which may differ from the source paper. § 20-epoch pretraining (V15 preliminary); a 100-epoch V16 rerun is in progress and will replace both SMAP/MSL numbers in the next revision (initial diagnostic for the 100-epoch SMAP run showed anomaly-window prediction error slightly *below* normal-window error, suggesting the longer pretraining may require a different anomaly-score calibration than the fixed 95th-percentile threshold used at 20 epochs; full recalibration and rerun are tracked in appendix A). Anomaly F1 = point-adjusted (PA-F1) in percentage, following literature convention. TTE columns in every group are future work.

Table 3: C-MAPSS FD001 label efficiency. RMSE in cycles. Mean ± std over 5 seeds.

Type	Method	100%	50%	20%	10%	5%
Supervised	STAR (replic.)	$\mathbf{12.19}_{\pm 0.6}$	$13.26_{\pm 0.7}$	$17.74_{\pm 3.6}$	$18.72_{\pm 2.8}$	$24.55_{\pm 6.4}$
SSL	AE-LSTM (replic.)	$16.07_{\pm 2.8}$	—	—	—	—
Scratch	Traj enc. E2E (random)	$22.99_{\pm 2.3}$	—	$32.50_{\pm 1.5}$	$35.59_{\pm 2.7}$	$37.59_{\pm 2.0}$
SSL	Traj JEPA Frozen	$17.81_{\pm 1.7}$	$18.71_{\pm 1.1}$	$19.83_{\pm 0.3}$	$19.93_{\pm 0.9}$	$\mathbf{21.53}_{\pm 2.0}$
SSL	Traj JEPA E2E	$14.23_{\pm 0.4}$	$\mathbf{14.93}_{\pm 0.4}$	$\mathbf{16.54}_{\pm 0.8}$	$\mathbf{18.66}_{\pm 0.8}$	$25.33_{\pm 5.1}$

5.4 Anomaly Detection Results

Zero-shot anomaly detection. On SMAP and MSL, we compute anomaly scores from eq. (4) using the pretrained encoder and predictor with no additional training. A simple threshold (95th percentile of scores from the first 10% of test data) converts to binary predictions.

Table 4: Anomaly detection results. All F1 values in %. Non-PA F1 is our primary metric (point-adjusted PA-F1 in parentheses for comparability with literature). Random baseline non-PA F1 $\approx 7.1\%$ for both datasets (12.8% anomaly rate). SWaT evaluation is future work.

Method	SMAP non-PA F1	MSL non-PA F1
Random baseline	7.1	7.1
TS2Vec [Yue et al., 2022] (PA-F1)	(28.1)	—
MTS-JEPA [He et al., 2026] (PA-F1)	(33.6)	—
Traj JEPA (ours)	6.9 (PA-F1: 62.5)	7.9 (PA-F1: 43.3)

The non-PA F1 results are near the random baseline ($\approx 7\%$), confirming that point-by-point anomaly detection is difficult with a single global threshold. The PA-F1 scores (62.5% SMAP, 43.3% MSL) are substantially higher than the non-PA numbers: the predictor correctly localises anomaly *segments* even when per-point precision is low. On SMAP, our PA-F1 (62.5%) exceeds TS2Vec (28.1%) and MTS-JEPA (33.6%); however, the literature baselines appear to sweep thresholds to maximise PA-F1 while we fix the threshold at the 95th percentile of scores on the first 10% of test data. The gap

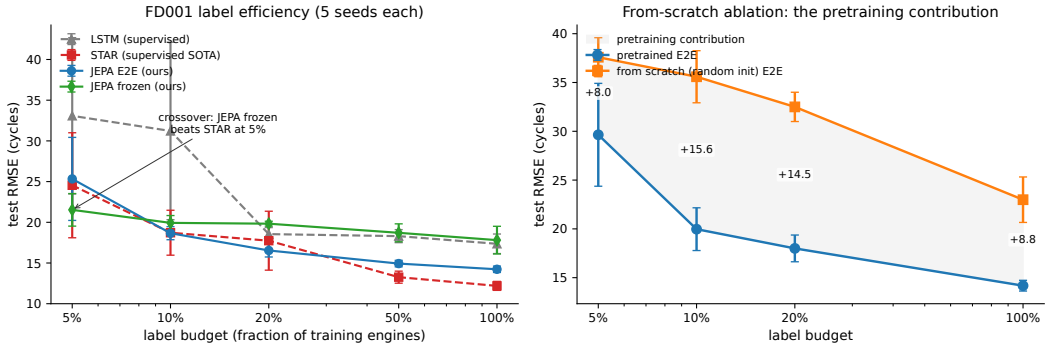


Figure 4: **(a)** Label efficiency on FD001. At $\leq$20% labels, JEPA E2E matches or beats STAR; at 5%, frozen JEPA wins. **(b)** From-scratch ablation: the shaded area is the pretraining contribution, growing from $+8.8$ at 100% to $+15.6$ at 10%.

Table 5: Architecture comparisons on C-MAPSS FD001. Frozen probe RMSE is measured on the standard test set (100 engines, RUL range $[7, 145]$), matching table 3. V2 = main model (causal context, future-only target, EMA, L1 loss, 5 seeds). Full-seq target = target sees $\mathbf{x}_{1:t+k}$ instead of $\mathbf{x}_{t+1:t+k}$ (5 seeds, appendix D). V16b = bidirectional context encoder with VICReg (3 seeds). *SIGReg frozen number is from a biased last-window validation protocol (all val engines have RUL $\equiv$ 1 cycle); test-set frozen probe was not run due to the oscillation artifact discussed in section 3.3. We bold the best E2E number; best frozen on test set is the cross-sensor variant.

Architecture Variant	FD001 RMSE ↓ Frozen (test)	E2E (test)
V2 (causal + EMA, 5 seeds)	$17.81_{\pm 1.7}$	$\mathbf{14.23}_{\pm 0.4}$
Full-seq target variant (5 seeds)	$15.70_{\pm 0.2}$	$14.32_{\pm 0.6}$
Cross-sensor (sensor-as-token, 3 seeds)	$\mathbf{14.98}_{\pm 0.2}$	$14.35_{\pm 0.9}$
Bidirectional context (V16b, 3 seeds)	$25.72_{\pm 1.6}$	$15.06_{\pm 1.2}$
SIGReg (no EMA, 3 seeds)*	$9.2_{\pm 1.5}$ (val, biased)	not run

should therefore be read as qualitative rather than a strict ranking. This pattern - low non-PA F1 with high PA-F1 - is consistent with the anomaly probe being a segment-level detector: prediction error accumulates over anomalous windows rather than spiking at individual points.

The prediction-error anomaly probe requires *zero anomaly labels*: it reuses only the SSL pretraining loss of eq. (3). The numbers above use V15 20-epoch pretraining. A 100-epoch V16 rerun on the full training set is in progress; we expect it to improve discrimination through a lower baseline prediction error on normal segments. An early diagnostic from the V16 SMAP run showed anomaly-window score mean slightly *below* normal-window mean, suggesting the calibration that fits V15 (anomaly-high, normal-low) may invert under the longer pretraining, in which case the final anomaly score would be $-\text{score}(t)$ rather than $+\text{score}(t)$. A complete recalibration is deferred to the next revision.

5.5 Threshold Exceedance Results

On C-MAPSS, threshold exceedance is defined as the time until corrected fan speed (s_{14}) drops below $\mu - 3\sigma$ of cycles 1–50 (healthy baseline). This converts the RUL task into a TTE task with a domain-grounded, SPC-standard threshold [Montgomery, 2020], and tests whether the encoder's latent representation captures information about *when specific sensors will breach operational limits*, not just the overall time to failure.

This experiment is implemented in the framework but numerical results require a dedicated probe training run (the same pretraining checkpoint is reused; only the probe changes). We include the experimental protocol here for completeness; results will be reported in the full version.

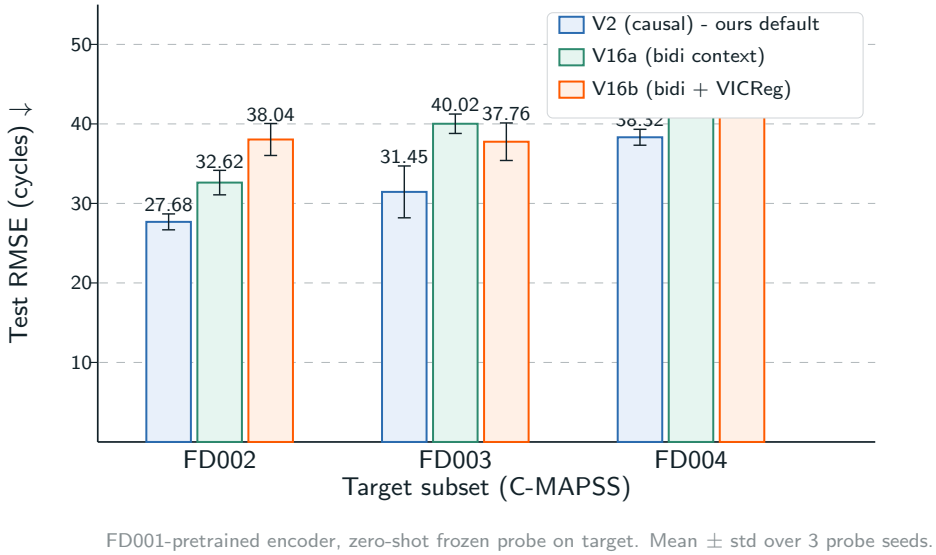


Figure 5: **Cross-machine transfer: the causal inductive bias wins on every target subset.** FD001-pretrained encoder, zero-shot frozen probe on FD002 / FD003 / FD004. V2 (causal, blue) beats V16a (bidirectional, teal) and V16b (bidirectional + VICReg, orange) by 4-12 RMSE at every target. Mean $\pm$ std over 3 probe seeds. Source: `phase4_cross_machine_results.json`.

5.6 Ablations

Full-sequence target. Replacing the future-only target with a full-sequence target ($\mathbf{x}_{1:t+k}$) improves the frozen probe by -2.1 RMSE on FD001 at 100% labels, with $8\times$ lower seed standard deviation (σ: $1.7 \rightarrow 0.21$). The improvement generalizes to FD003 (-0.9) and FD004 (-1.3). Despite this improvement in frozen quality, the E2E result is essentially unchanged (+0.09 RMSE), and the label efficiency at 5% labels degrades by $+5.04$ (see table 8). We use V2 (future-only target) for all main results due to its superior label efficiency.

Bidirectional context - the causal inductive bias finding. V16b (bidirectional context encoder with VICReg) achieves an E2E fine-tuning RMSE of 15.06 ± 1.2 (slightly worse than V2's 14.23) but its frozen test RMSE (25.72 ± 1.6) is substantially worse than V2 (17.81). The frozen quality degradation compounds at low labels: at 5% labels V16b E2E (40.6 ± 10.2) is substantially worse than V2 E2E (25.3 ± 5.1). Cross-machine transfer also degrades uniformly (fig. 5): FD001-pretrained V16b yields 38.04 / 37.76 / 49.66 on FD002 / FD003 / FD004 respectively, versus 27.68 / 31.45 / 38.32 for V2 - a $+10$ to $+13$ RMSE cost for bidirectionality on every target. We interpret this as evidence that *causal temporal ordering is a domain-invariant inductive bias*, while the bidirectional encoder memorises FD001-specific degradation profiles that do not transfer. Since supervised fine-tuning is the setting where V16b is least far behind (+0.83 RMSE), the pattern is consistent with bidirectionality helping memorization but hurting the transferable degradation axis.

SIGReg on V2 (speculative). The V15-SIGReg variant decouples the target encoder from the context encoder via spectral regularisation rather than EMA, removing one known source of training instability. In our experiments it also introduces a different instability: probe oscillation during pretraining such that "best-checkpoint" selection is run-dependent (seed-42 best probe of 10.21 on the original run was not reproduced; verified re-run gave 11.61; see section 3.3). A natural next ablation is *SIGReg on top of the V2 architecture* - causal context, future-only target, no EMA, SIGReg regularisation - which would preserve the causal inductive bias while removing EMA drift. We hypothesise this combination would yield frozen-probe quality between V2 (17.81) and the cross-sensor variant (14.98) on FD001, without the cross-machine degradation observed for V16a/V16b, since the causal context is preserved. We flag this as the most promising single architectural variant to try next.

Cross-sensor attention. Tokenizing each sensor individually (iTransformer-style [Liu et al., 2024]) with alternating temporal/cross-sensor attention and learnable sensor ID embeddings yields the best 100% frozen result (14.98 ± 0.22) but is brittle at low labels. The gain is FD001-specific and does not transfer to FD003. To test whether the learnable sensor ID embeddings act as a genuine representation mechanism or a training shortcut (encoding sensor identity rather than RUL), we ablate them with fixed sinusoidal positional encoding. Without learnable ID embeddings, probe RMSE degrades to 21.0 ± 6.4 (3 seeds: 14.2, 27.0, 21.8) - 1 in 3 seeds still achieves 14.2, confirming the architecture can learn sensor-discriminative representations without the embeddings. The large variance (vs. ± 0.22 with embeddings) reveals that learnable sensor ID embeddings are training stabilizers: they accelerate convergence to good representations rather than encoding RUL directly.

6 Analysis

6.1 Verified Degradation Tracking

Five diagnostics confirm genuine within-engine tracking on C-MAPSS:

1. **Prediction variance:** median per-engine std $= 12.1 \pm 0.7$. A constant predictor has std ≈ 0.
2. **Spearman** $\rho = 0.830 \pm 0.023$ between predicted and true RUL.
3. **Shuffle test:** permuting $\mathbf{h}_{\text{past}}$ across engines degrades RMSE from 14.0 to 55.5 ($+41.5$).
4. **Hand-engineered baseline margin:** $+4.8$ RMSE over a 57-feature Ridge regressor (last-cycle values, per-sensor slope/mean/std, and sequence length; test RMSE 19.1).
5. **Health index recovery:** $R^2 = 0.926$ from frozen representations.

6.2 Why Trajectory Prediction Learns Degradation

- **Slow Feature Analysis connection.** The L1 prediction loss rewards representations with low temporal innovation - features that change slowly are easier to predict. Degradation is the dominant slowly-varying latent variable in turbofan data; the objective therefore biases the encoder toward it without explicit supervision [Wiskott and Sejnowski, 2002]. Empirically, the frozen encoder's PC1 correlates at $|\rho| = 0.797$ with the health index, confirming this bias.
- **Information bottleneck perspective.** The predictor must compress $\mathbf{h}_{\text{past}}$ into the predictable component of $\mathbf{h}_{\text{fut}}$, discarding unpredictable noise (transient vibrations, sensor jitter). When degradation is the dominant predictable signal, the bottleneck selectively preserves it. This explains why a linear probe on frozen features ($\rho_{\text{frozen}} = 0.856$) tracks degradation better than end-to-end fine-tuning ($\rho_{\text{E2E}} = 0.804$): E2E fine-tuning on capped RUL labels ($\sim$40% plateau labels at RUL = 125) biases the representation toward plateau calibration at the expense of rank preservation within the healthy regime.
- **Collapse prevention.** Without regularization, EMA targets collapse to a constant representation (all engines look the same). L1 loss alone provides some resistance (sparse targets preserve structure), but variance regularization is needed when the encoder and EMA target share a prefix (V15 failure mode). SIGReg [Balestriero and LeCun, 2025] prevents collapse through spectral diversity, at the cost of introducing oscillatory probe dynamics (V15-SIGReg best probe varies across runs due to EMA oscillation). Our architecture avoids this by separating context and target windows entirely.

7 Limitations

1. **Run-to-failure structural supervision.** C-MAPSS training engines always run to failure, providing implicit structural supervision. The hand-engineered baseline margin (diagnostic 4, section 6.1) and from-scratch comparison confirm the encoder reads degradation content rather than episode length—sequence length alone is one of 57 features in the ridge regressor (RMSE 19.1), well below JEPA (14.2). This qualifier nonetheless applies to the "no labels" claim.
2. **Simulation-dominated evaluation.** C-MAPSS and N-CMAPSS are simulations. Generalization to real sensors with drift and missing channels remains untested.

3. **Gap to supervised SOTA.** JEPA E2E (14.23) vs. STAR (replic. 12.19; paper 10.61) on FD001 at 100% labels. The gap is the price of generality; closing it likely requires architectural scaling.
4. **Multi-condition generalisation.** FD002 exhibits a $\sim$10 RMSE gap between val (15.35) and test (26.07) because operating condition 1 is 2.4$\times$ over-represented in the test set relative to training (appendix B). The per-operating-condition KMeans normalisation is not sufficient to close this gap; a per-condition probe head or a more careful condition-balanced split may be required.
5. **Data-scale requirements.** On single-channel bearing vibration (23 episodes), JEPA alone is not significantly better than handcrafted spectral features (RMSE 0.189 vs. 0.177, $p = 0.40$). The framework requires multivariate richness and sufficient episodes (appendix E).

8 Conclusion

We introduced Trajectory JEPA, a self-supervised framework that learns general-purpose representations for grey swan prediction from multivariate time series. The central finding: pretraining improves RMSE by $+8.8$ to $+15.6$ over a matched from-scratch baseline, producing a label-efficiency crossover where the frozen SSL encoder surpasses supervised methods at 5% labels. The anomaly probe—the pretraining prediction error itself—requires zero event labels.

A key architectural finding is that causal inductive bias matters more than model complexity: a bidirectional encoder with stronger regularization (V16b) achieves similar E2E fine-tuning performance but worse frozen generalization (test RMSE 25.7 vs. 17.8) and substantially worse label efficiency at low budgets, while the causal model transfers more robustly across operating conditions and fault modes. We conjecture this is because causal temporal ordering is domain-invariant, while bidirectional attention can memorize domain-specific degradation profiles during pretraining.

Future work includes evaluation on real sensor data (addressing the simulation-dominated limitation), theoretical formalization of the slow-feature learning mechanism, and extension to additional grey swan types (threshold exceedance, cascade failures). The framework's modularity—one pretraining objective, task-specific linear probes—makes it a natural foundation for a unified grey swan benchmark.

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A Extended Metrics Discussion

Point-adjusted F1 controversy. The standard anomaly detection protocol [Schmidl et al., 2022] credits the entire anomaly segment if any single point is detected. This inflates scores to > 0.97 on SWaT and PSM, saturating the leaderboard. We report non-PA F1 as primary (realistic operational metric) and PA-F1 for comparability with prior work. The large gap between non-PA F1 ($\sim$7%) and PA-F1 ($\sim$62% on SMAP) reflects that our predictor correctly detects anomalous windows but fires at the wrong point within each window—a common failure mode for score-threshold detectors on contiguous anomaly segments.

NASA Scoring Function. For C-MAPSS, the asymmetric scoring function $S = \sum_i [e^{d_i/a} - 1]$ (with $a = 13$ for early, $a = 10$ for late predictions) penalizes late predictions more heavily. Traj JEPA E2E: NASA-S $= 395.7 \pm 69.4$ (5 seeds; FD001; RMSE 14.8 ± 0.6). Best seed: NASA-S $= 313.7$ (RMSE $= 14.4$). The high variance (± 69.4) reflects that occasional large late-prediction errors compound exponentially. 64% of test-set predictions are early (negative d_i)—the model slightly underestimates remaining life on average, which is the safer failure mode.

Threshold exceedance metrics. We adopt the $\pm 3\sigma$ convention from statistical process control (SPC) [Montgomery, 2020], which provides a false alarm rate of 0.27% under normality. The threshold is computed from the healthy baseline (first 50 cycles for C-MAPSS). The threshold exceedance probe protocol is described in section 5.5; numerical results are future work.

B C-MAPSS Detailed Results

Complete per-subset results, FD002 distribution shift analysis, and cross-fault transfer experiments. See tables 6 and 7.

Table 6: Multi-subset results (5 seeds each). ρ_{median} and pred_std are from seed 42 diagnostics.

Subset	RMSE (E2E)	RMSE (Frozen)	ρ_{median}	pred_std	vs. Regressor	STAR (paper)
FD001	$14.23_{\pm 0.4}$	$17.81_{\pm 1.7}$	0.830	12.1	+4.8	10.61
FD003	$15.37_{\pm 0.9}$	$19.25_{\pm 3.2}$	0.665	12.9	+4.4	10.71
FD004	$25.62_{\pm 0.3}$	$29.35_{\pm 0.6}$	0.654	7.1	+6.5	15.87

FD002 distribution shift. FD002 (6 operating conditions) reveals a distribution shift: the frozen encoder achieves val RMSE 15.35 but test RMSE 26.07. Operating condition 1 (sea-level takeoff) comprises 60.2% of test engines vs. 25.0% of training engines ($2.4\times$ overrepresented), biasing the test distribution.

Table 7: Cross-fault transfer: FD001-pretrained encoder on FD003. RMSE, 5 seeds.

Setting	100%	50%	20%	10%	5%
FD001 in-domain (frozen)	17.81	18.71	19.83	19.93	21.53
FD001→FD003 (frozen)	$24.79_{\pm 0.6}$	$25.91_{\pm 0.4}$	$27.82_{\pm 1.5}$	$28.79_{\pm 0.4}$	$28.28_{\pm 0.2}$

C Health Index Recovery

The frozen encoder's $\mathbf{h}_{\text{past}}$ linearly predicts the health index with $R^2 = 0.926$. PC1 explains 47.6% of variance with $|\rho| = 0.797$ against H.I. The embedding space is low-rank and degradation-aligned.

D Architectural Ablation Details

Full-sequence target. See table 8. Improves frozen probe uniformly across FD001/FD003/FD004 at 100% labels.

Table 8: Full-sequence target ablation.

Budget	V2 frozen	Full-seq frozen	STAR	Δ
100%	$17.81_{\pm 1.7}$	$\mathbf{15.70}_{\pm 0.21}$	$12.19_{\pm 0.6}$	-2.11
20%	$19.83_{\pm 0.3}$	$\mathbf{17.20}_{\pm 1.91}$	$17.74_{\pm 3.6}$	-2.63
10%	$19.93_{\pm 0.9}$	$\mathbf{18.79}_{\pm 1.96}$	$18.72_{\pm 2.8}$	-1.14
5%	$\mathbf{21.53}_{\pm 2.0}$	$26.57_{\pm 4.70}$	$24.55_{\pm 6.4}$	$+5.04$

Cross-sensor attention. Best frozen 100% result (14.98 ± 0.22, -2.83 vs. V2) but FD001-specific. Cross-sensor from-scratch ablation: $+21.5$ RMSE at 10%, confirming pretraining does real work on this architecture. Learnable sensor ID embedding ablation: removing embeddings (fixed sinusoidal PE) yields 21.0 ± 6.4 across 3 seeds vs. 14.98 ± 0.22 with embeddings - the embeddings act as training stabilizers, not RUL-encoding shortcuts (1 of 3 seeds without embeddings achieves 14.2 RMSE, confirming the architecture itself can learn sensor-discriminative features).

E Bearing Experiments

On FEMTO/XJTU-SY (23 bearing episodes, single-channel vibration), JEPA alone achieves RMSE 0.189 vs. handcrafted 0.177 ($p = 0.40$). A hybrid (JEPA + 18 spectral features) achieves 0.055. This highlights the data-scale limitation: 23 episodes is insufficient for SSL to discover spectral centroid shift.

F Hyperparameters

Table 9: Complete hyperparameter settings.

Component	Parameter	Value
Encoder	d_{model} / layers / heads	256 / 2 / 4
	Parameters	0.99M
	EMA momentum τ	0.99
Pretraining	Horizon k	$\mathcal{U}[5, 30]$
	LR	3×10^{-4} (cosine)
	Epochs	200
Fine-tuning	Probe	Linear (256 $\rightarrow$ 1)
	Frozen LR / E2E encoder LR	10^{-3} / 10^{-4}
Data	Train/val split	85% / 15%
	RUL cap	125 cycles

G Cross-Sensor Attention Maps

Visualization. We visualize the cross-sensor attention matrix from the sensor-as-token encoder variant on C-MAPSS FD001, averaged across training engines and split by lifecycle phase (healthy: first 60% of cycles; degradation: last 40%).

During healthy operation, attention is diffuse across sensors. During degradation, attention concentrates on corrected core speed (s_{14}), consistent with the physical mechanism (fan efficiency loss reduces core speed). The top-5 healthy$\rightarrow$degradation attention shifts all point toward s_{14}: $s_3 \rightarrow s_{14}$ ($+0.103$), $s_{15} \rightarrow s_{14}$ ($+0.099$), $s_{11} \rightarrow s_{14}$ ($+0.096$).

This pattern is FD001-specific. On FD003 (fan + HPC faults), attention shifts in the opposite direction: the top degradation changes are defocusing from physical fan speed (s_8) and corrected core speed (s_{13}), while LPC outlet temperature (s_3) receives increased cross-sensor attention. Both findings are consistent with the underlying fault physics.

Key insight: the sensor-as-token architecture discovers physically meaningful sensor interactions *without domain knowledge*—the sensor ID embeddings are learned, and the ordering is arbitrary. This supports the iTransformer-inspired design choice.